

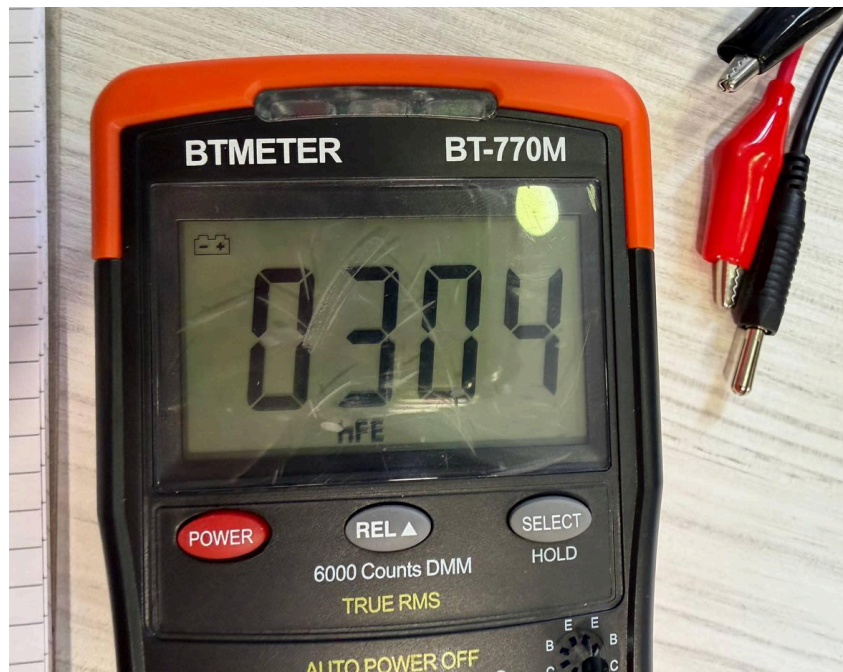
BJT Amplifier

ENGS152 Circuits Lab

10 April 2026

Step 1: h_{FE} (β) value investigation

Before building the circuit, we are asked to verify our NPN 2N3904 transistor, to ensure that gain matches our design assumptions. To do so multimeter's h_{FE} setting has been used. The transistor leads have been inserted into the corresponding holes in the h_{FE} test input jack, and the value has been recorded.



As we can see the h_{FE} (β) value is 304. Which almost falls into the practical value range of 150-300.

Step 2: Circuit Design and DC Bias Verification

Now, we are asked to build the DC bias network for the BJT amplifier and verify the quiescent operating point (Q-point).

We are given the following design requirements and assumptions, which we should consider throughout the circuit design:

1. $V_{cc} = 10\text{ V}$
2. β or $hFE = 304$
3. $I_c = 2\text{ mA}$
4. $V_c = \frac{V_{cc}}{2}$
5. $V_E = \frac{V_{cc}}{10}$
6. $I_E = I_c$
7. $I_{BIAS} > 10 * I_B$
8. $V_{BE} = 0.7\text{ V}$
9. $V_T = 25\text{ mV}$
10. **2N3904** transistors to be used

Pre-Lab calculations were used to determine the resistor values for R_1 , R_2 , R_L , R_E

$$R_L = \frac{V_{cc} - V_c}{I_c} = \frac{10V - 5V}{2mA} = 2.5\text{ k}\Omega$$

$$R_E = \frac{V_E}{I_E} = \frac{1V}{2mA} = 500\ \Omega$$

$$V_B = V_E + V_{BE} = 1 + 0.7 = 1.7\text{ V}$$

$$I_B = \frac{I_c}{\beta} = \frac{0.002}{304} = 6.5\ \mu A$$

$$I_{BIAS} = 10 \times I_B = 65\ \mu A$$

$$I_{BIAS} = 65 \mu A > 10 * I_B$$

$$R_2 = \frac{V_B}{I_{BIAS}} = \frac{1.7}{65 \cdot 10^{-6}} = 26.15 \text{ k}\Omega$$

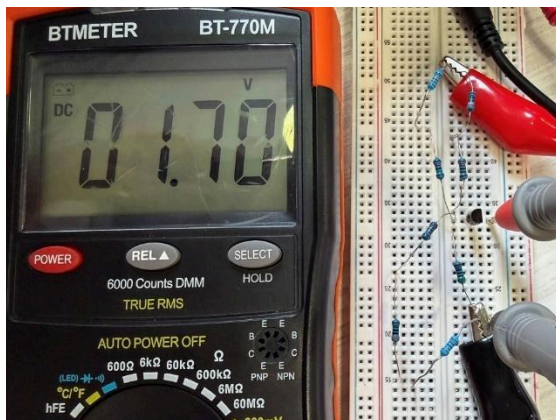
$$R_1 = \frac{V_{CC} - V_B}{I_{BIAS} + I_B} = \frac{10 - 1.7}{71.5 \cdot 10^{-6}} = 116 \text{ k}\Omega$$

After calculating the resistor values, we are asked to choose the nearest standard resistor values available in the lab. In order to ensure the calculations stay relatively the same our team decided to use resistors in series to better match the values.

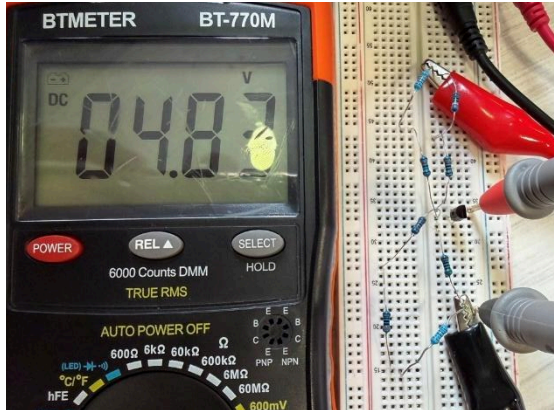
	Actual Value	Selected Value
R_1	116 k Ω	100 k Ω + 15 k Ω
R_2	26.15 k Ω	22 k Ω + 3 k Ω + 1 k Ω
R_L	2.5 k Ω	2.2 k Ω + 330 Ω
R_E	500 Ω	510 Ω

After finding the resistor values we are asked to assemble the circuit without adding the capacitors yet, with only the DC power supply (VCC) connected. A multimeter has been used to verify the Quiescent (Q-point) conditions.

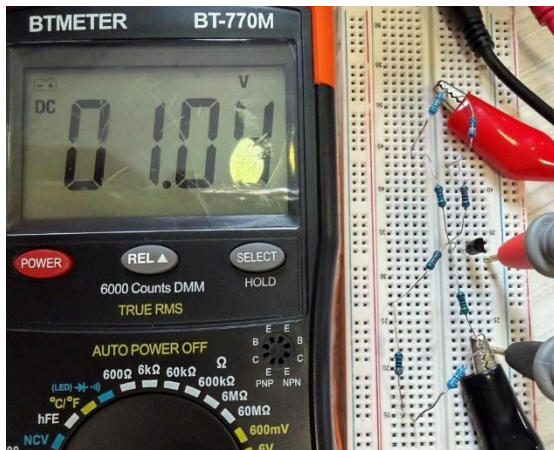
BASE VOLTAGE



COLLECTOR VOLTAGE



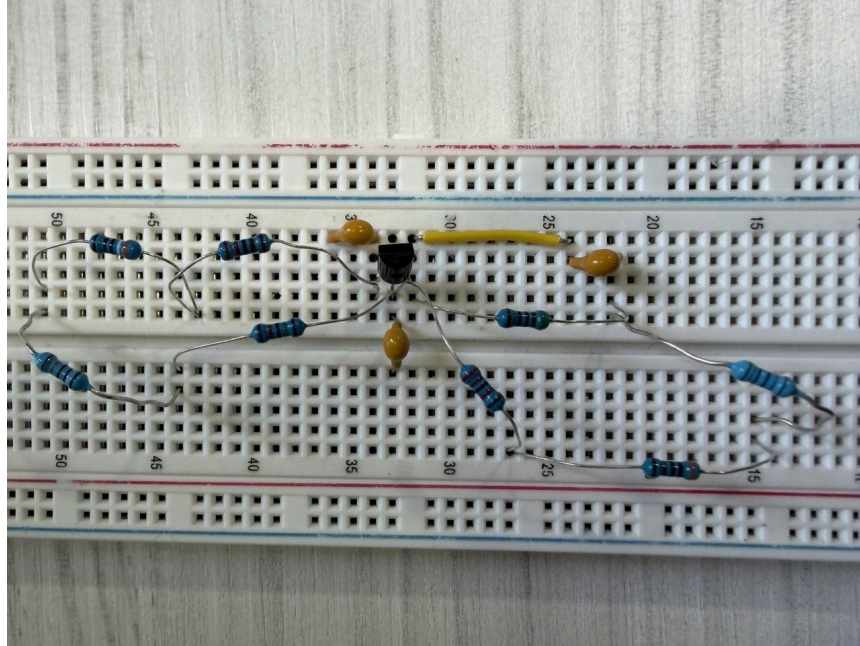
EMITTER VOLTAGE



As we can see the measured values beautifully match the theoretical values within a very low error percentage. The small differences within the measured and theoretical values can be explained by the limited selection of the resistors in the lab and the even smallest differences can be explained by internal resistance of the components.

Once the DC operating point is verified, we are asked to add the capacitors. They will allow AC signals to pass while blocking the DC components.

$$C_1 = C_2 = C_3 = 10\mu F$$



Once the capacitors are inserted in the circuit, we can proceed to the theoretical cutoff frequency calculations.

$$r_e = \frac{V_T}{I_E} = \frac{25 \times 10^{-3}}{2 \times 10^{-3}} = 12.5 \, \Omega$$

$$\beta r_e = 305 \times 12.5 = 3812.5 \, \Omega$$

$$R_{in} = \left(\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{\beta r_e} \right)^{-1} = \left(\frac{1}{116 \times 10^3} + \frac{1}{26 \times 10^3} + \frac{1}{3812.5} \right)^{-1} \approx 3223.4$$

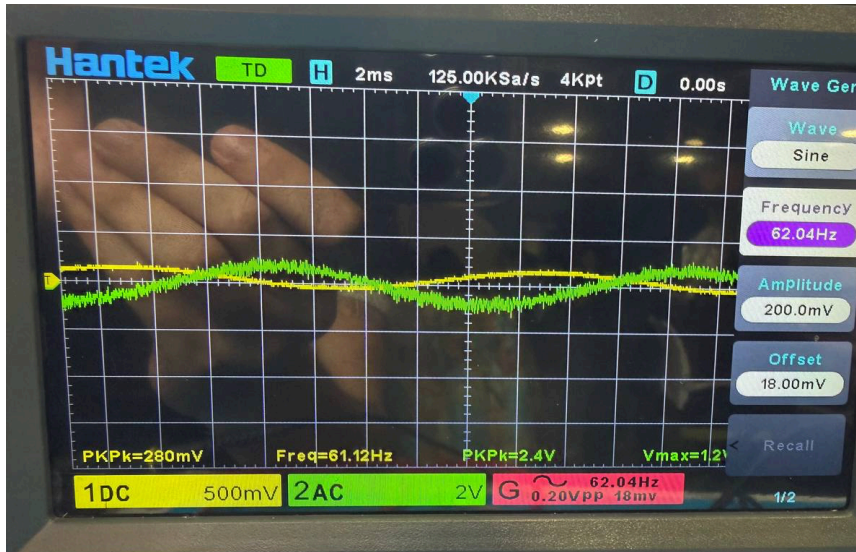
$$f_c = \frac{1}{2\pi R_{in} C_1} = 4.94 \, \text{Hz}$$

Step 3: AC Analysis (Cut-off Frequencies)

In this part of the lab, we are asked to connect the signal generator to the amplifier input. The input must be a sine wave with an amplitude of 200mV.

We tested the amplifier's performance across two different frequency ranges relative to our calculated cutoff frequency (f_c):

1. Close to the cutoff frequency



Input PkPk: 280 mV

Input Frequency: 61.12 Hz

Output PkPk: 2.4V

Output Frequency: We actually didn't notice that we didn't set up the channel 2 frequency display, so it's missing. We sincerely apologise.

$V_{out}/V_{in} = \text{gain} = 8.57$

2. Much larger than the cutoff frequency ($f \gg f_c$)



Input PkPk: 260 mV

Input Frequency: 661.38 Hz

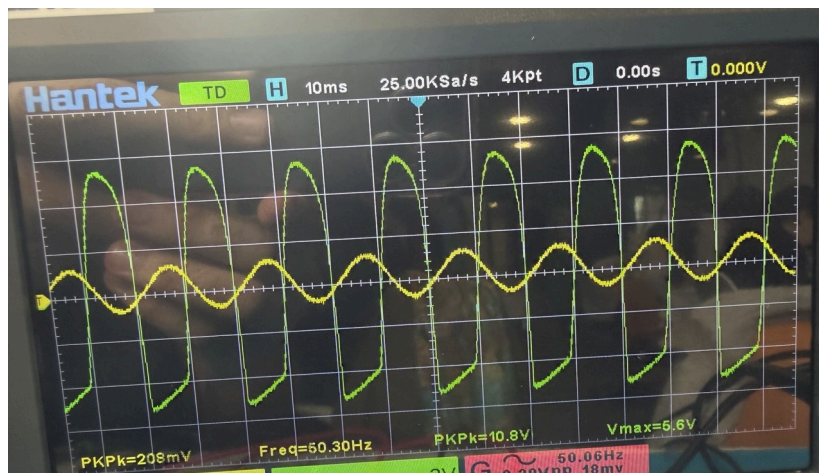
Output PkPk: 8.88 V

$V_{out}/V_{in} = \text{gain} = 34.15$

As we may notice, when we increase the input frequency, the output is greatly amplified. This can be noted also when we look at the gain, which is much less when the input frequency is close to the cutoff frequency compared to the $f \gg f_c$ case.

Then we investigated how the emitter resistance R_E directly controls the amplifier's gain by replacing the fixed emitter resistor R_E with a potentiometer of 1k Ohms.

In the first picture we observe the situation when $R=0$ and we increase it gradually until $R=R_E$. When the resistance is lower than R_E we notice big gain and an unstable output signal waveform (clipped), however when the resistance value is close to R_E , the gain is close to 1 and we have a stable sine output waveform. In the last picture $R > R_E$, which results in negative gain but a clearer sine wave, which is the most stable response.





Next we increased the input amplitude(V_{IN}) gradually. At $V=980$ mV we noticed slight output signal clipping. The signal clipping happens when the input voltage is

increased enough that the transistor is forced out of its linear or active region and into either cutoff or saturation. It's best to choose V_C to be 50% of the V_{CC} because the output can swing equally up and down without hitting cutoff or saturation which means maximum undistorted signal.

